

# Brendan M. O'Connor

MCWILLIAMS FELLOW · CARNEGIE MELLON UNIVERSITY

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## Professional Experience

- Sep. 2023 - present **McWilliams Postdoctoral Fellow**, Carnegie Mellon University, Department of Physics & The McWilliams Center for Cosmology and Astrophysics
- 2019 - 2023 **Faculty Assistant**, Department of Astronomy, University of Maryland, College Park
- 2019 - 2023 **Research Assistant**, Astrophysics Science Division, NASA Goddard Space Flight Center (GSFC)

## Education

### The George Washington University

PHD PHYSICS – GPA: 4.0

- Advisor: Dr. Chryssa Kouveliotou; Co-Advisors: Eleonora Troja and Brad Cenko
- Thesis: The Transient Universe: Compact Objects Near and Far

Washington, DC

August 2017 - August 2023

### The George Washington University

MPHIL PHYSICS – GPA: 4.0

Washington, DC

August 2017 - May 2020

### The George Washington University

MS PHYSICS – GPA: 4.0

Washington, DC

August 2017 - January 2020

### Union College

BS PHYSICS – GPA: 3.8 (*summa cum laude*)

- Minor in Astrophysics
- Thesis: Colliding Wind Binaries with Orbital Motion; Advisor: Francis Wilkin

Schenectady, NY

August 2013 - June 2017

## Research Interests

- Gamma-ray bursts and their host galaxies and environments
- Gravitational waves and multi-messenger astrophysics
- Neutron star mergers and kilonovae
- Fast Radio Bursts and their host galaxies and environments
- Galactic X-ray transients: magnetars, cataclysmic variables, high-mass X-ray binaries
- Surveys and serendipitous optical, infrared, or high-energy transients

My research has mainly focused on using the following telescopes and observatories:

- **X-rays:** *Swift/XRT, NICER, NuSTAR, Chandra, XMM-Newton*
- **Ultraviolet/optical/infrared:** *Hubble Space Telescope (HST), James Webb Space Telescope (JWST), Swift/UVOT, Gemini, Keck, Dark Energy Camera (DECam), Southern African Large Telescope (SALT), Lowell Discovery Telescope (LDT)*

## Publications

**ORCID:** <https://orcid.org/0000-0002-9700-0036>

As of November 15, 2024: I have published 10 first-author, 6 second-author, and 4 third-author publications. My first author publications have > 200 citations. I have authored or co-authored 41 publications which have > 1000 citations. My

h-index is 16. In addition to these refereed publications I have led 20+ GCN Circulars and Astronomer's Telegrams.

#### IN PREPARATION (SORTED BY POSITION)

\*Undergraduate Student \*\*Graduate Student

1. Srinivasaragavan, G. \*\*, Perley, D., Ho, A., **O'Connor, B.**, et al., 2024. *Multi-wavelength Analysis of an Optically-Discovered Afterglow Without a Detected Gamma-ray Counterpart*. To be submitted.

#### STUDENT LED (SORTED BY POSITION)

\*Undergraduate Student \*\*Graduate Student

1. Kaur, R. \*, **O'Connor, B.**, Palmese, A., Kunnumkai, K. 2024. *Detecting prompt and afterglow jet emission of gravitational wave events from LIGO/Virgo/KAGRA and next generation detectors*. Submitted to PRD. doi: 10.48550/arXiv.2410.10579
2. Freeburn, J. \*\*, **O'Connor, B.**, Cooke, J., Dobie, D., et al., 2024. *GRB 220831A: a hostless, intermediate Gamma-ray burst with an unusual optical afterglow*. Submitted to MNRAS. doi:10.48550/arXiv:2411.14749
3. Cabrera, T. \*\*, Palmese, A., Hu, L., **O'Connor, B.**, et al., 2024. *Searching for electromagnetic emission in an AGN from the gravitational wave binary black hole merger candidate S230922g*. Submitted to ApJ. doi: 10.48550/arXiv.2407.10698

#### FIRST AUTHOR (SORTED BY YEAR)

1. **O'Connor, B.**, Pasham, D., Andreoni, I., Hare, J., Beniamini, P., et al., 2024. *Characterization of a peculiar Einstein Probe transient EP240408a: an exotic gamma-ray burst or an abnormal jetted tidal disruption event?* Submitted to ApJL. doi:10.48550/arXiv.2410.21622
2. **O'Connor, B.**, Beniamini, P., and Gill, R., 2024. *X-ray Afterglow limits on the viewing angles of short gamma-ray bursts*. MNRAS, 533, 1629.
3. **O'Connor, B.**, Kouveliotou, C., Evans, P. A., Gorgone, N., van Kooten, A. J., et al., 2023. *The Swift Deep Galactic Plane Survey (DGPS) Phase-I Catalog*. ApJS, 269, 49. [VizieR/J/ApJS/269/49](https://ui.adsabs.org/2023ApJS...269...49O)
4. **O'Connor, B.**, Brink, J., Buckley, D. A. H., Mukai, K., Kouveliotou, C., et al., 2023. *Identification of 1RXS J165424.6-433758 as a Polar Cataclysmic Variable*. ApJ, 957, 89.
5. **O'Connor, B.**, Göğüş, E., Hare, J., Mukai, K., Huppenkothen, D., et al., 2023. *Swift Deep Galactic Plane Survey classification of Swift J170800-402551.8 as a candidate intermediate polar cataclysmic variable*. MNRAS, 525, 5015
6. **O'Connor, B.**, Troja, E., Ryan, G., Beniamini, P., et al., 2023. *A structured jet explains the extreme GRB 221009A*. **Science Advances**, 9, eadi1405. [10.1126/sciadv.adi1405](https://doi.org/10.1126/sciadv.adi1405)
7. **O'Connor, B.**, Troja, E., Dichiara, S., Beniamini, P., et al., 2022. *A deep survey of short GRB host galaxies over  $z \sim 0 - 2$ : implications for offsets, redshifts, and environments*. MNRAS, 485, 4890
8. **O'Connor, B.**, Göğüş, E., Huppenkothen, D., Kouveliotou, C., et al., 2021. *Identification of an X-Ray Pulsar in the BeXRB System IGR J18219-1347*. ApJ, 927, 139
9. **O'Connor, B.**, Troja, E., Dichiara, S., Chase, E. A., et al., 2021. *A tale of two mergers: constraints on kilonova detection in two short GRBs at  $z \sim 0.5$* . MNRAS, 502, 1279
10. **O'Connor, B.**, Beniamini, P., & Kouveliotou, C. 2020. *Constraints on the circumburst environments of short gamma-ray bursts*. MNRAS, 495, 4782

#### OTHER CO-AUTHOR (SORTED BY POSITION)

1. Srinivasaragavan, G., **O'Connor, B.**, Cenko, S. B., et al., 2023. *A Sensitive Search for Supernova Emission Associated with the Extremely Energetic and Nearby GRB 221009A*. ApJL, 949, L39.
2. Chase, E. A., **O'Connor, B.**, Fryer, C. L., Troja, E., et al., 2022. *Kilonova Detectability with Wide-field Instruments*. ApJ, 927, 163.
3. Troja, E., **O'Connor, B.**, Ryan, G., Piro, L., et al., 2022. *Accurate flux calibration of GW170817: is the X-ray counterpart on the rise?*. MNRAS, 510, 1902.
4. Bruni, G., **O'Connor, B.**, Matsumoto, T., Troja, E., et al., 2021. *Late-time radio observations of the short GRB 200522A: constraints on the magnetar model*. MNRAS, 505, L41.

5. Srinivasaragavan, G., Swain, V., **O'Connor, B.**, Anand, S., et al., 2023. *Characterization of SN 2023pel Associated with GRB 230812B: An Energetic GRB with an Ordinary Broad-lined Type Ic Supernova*. ApJL, 960, L18.
6. Yang, Y.-H., Troja, E., **O'Connor, B.**, Fryer, C. L., et al., 2023. *A lanthanide-rich kilonova in the aftermath of a long gamma-ray burst*. **Nature**, 626, 742.
7. Troja, E., Fryer, C. L., **O'Connor, B. (corresponding author)** Ryan, G., et al., 2022. *A long gamma-ray burst from a stellar merger in the nearby Universe*. **Nature**, 612, 228.
8. Dichiara, S., Troja, E., **O'Connor, B.**, Marshall, F. E., et al., 2020. *Short gamma-ray bursts within 200 Mpc*. MNRAS, 492, 5011.
9. Piro, L., Bruni, G., Troja, E., **O'Connor, B.**, et al., 2021. *The fast radio burst FRB 20201124A in a star-forming region: Constraints to the progenitor and multiwavelength counterparts*. A&A, 656, L15.
10. Dichiara, S., Troja, E., Beniamini, P., **O'Connor, B.**, et al., 2021. *Evidence of Extended Emission in GRB 181123B and Other High-redshift Short GRBs*. ApJL, 911, L28.
11. Becerra, R. L., Troja, E., Watson, A., **O'Connor, B.**, et al., 2023. *Deciphering the unusual stellar progenitor of GRB 210704A*. MNRAS, 522, 5204.
12. Ryan, G., van Eerten, H., Troja, E., Piro, L., **O'Connor, B.**, Ricci, R., 2024. *Modelling of Long-Term Afterglow Counterparts to Gravitational Wave Events: The Full View of GRB 170817A*. ApJ, 975, 131.
13. Gillanders, J. H., Troja, E., Fryer, C. L., Ristic, M., **O'Connor, B.**, et al., 2023, *Heavy element nucleosynthesis associated with a gamma-ray burst*. doi:10.48550/arXiv.2308.00633
14. Hammerstein, E., Cenko, S. B., Gezari, S., Veilleux, S., **O'Connor, B.**, et al., 2023. *Integral Field Spectroscopy of 13 Tidal Disruption Event Hosts from the ZTF Survey*. ApJ, 957, 86.
15. Ricci, R., Troja, E., Bruni, G., Matsumoto, T., Piro, L., **O'Connor, B.**, et al., 2021. *Searching for the radio remnants of short-duration gamma-ray bursts*. MNRAS, 500, 1708.
16. Enoto, T., Ng, M., Hu, C.-P., Güver, T., Jaisawal G. K., **O'Connor, B.**, et al., 2021. *A Month of Monitoring the New Magnetar Swift J1555.2—5402 during an X-Ray Outburst*. ApJL, 920, L4.
17. Troja, E., van Eerten, H., Zhang, B., Ryan, G., Piro, L., Ricci, R., **O'Connor, B.**, et al., 2020. *A thousand days after the merger: Continued X-ray emission from GW170817*. MNRAS, 498, 5643.
18. Kunnumkai, K.\*\*, Palmese, A., Farah A., et al., 2024. *Detecting electromagnetic counterparts to LIGO/Virgo/KAGRA gravitational wave events with DECam: Neutron Star Mergers*. Submitted to PRD. doi: 10.48550/arXiv:2411.13673
19. A. Trigg, R. Stewart, A. van Kooten, E. Burns, O. Roberts, et al., 2024. *Extragalactic Magnetar Giant Flare GRB 231115A: Insights from Fermi/GBM Observations*. Submitted to A&A. doi:10.48550/arXiv.2409.06056
20. Y. Yao, M. Guolo, F. Tombesi, R. Li, S. Gezari, et al., 2024. *Sub-relativistic Outflow and Hours-Timescale Large-amplitude X-ray Dips during Super-Eddington Accretion onto a Low-mass Massive Black Hole in the Tidal Disruption Event AT2022lri*. Submitted to ApJ. doi:10.48550/arXiv.2405.11343
21. Das K., Fremling C., Kasliwal M., Schulze S., Sollerman J., Karambelkar V., et al., 2024. *SN 2023zaw: an ultra-stripped, nickel-poor supernova from a low-mass progenitor*. ApJ, 969, L11.
22. Bruni, G., Piro, L., Yang, Y.-P., Quai, S., Zhang B., et al., 2024. *A nebular origin for the persistent radio emission of fast radio bursts*. **Nature**. <https://doi.org/10.1038/s41586-024-07782-6>
23. Ghosh, R., Laha, S., Meyer, E., et al., 2023. *Are-emerging bright soft-X-ray state of the changing-look Active Galactic Nucleus 1ES 1927+654: a multi-wavelength view*. ApJ, 955, 3.
24. Dichiara S., Troja E., Lipunov V., Ricci R., et al., 2022. *The early afterglow of GRB 190829A*. MNRAS, 512, 2337.
25. Dichiara, S., Becerra, R. L., Chase, E. A., Troja, E., et al., 2021. *Constraints on the Electromagnetic Counterpart of the Neutron-star-Black-hole Merger GW200115*. ApJL, 923, L32.
26. Gorgone, N. M., Woudt, P. A., Buckley, D., Mukai, K., et al., 2021. *Swift/XRT Deep Galactic Plane Survey Discovery of a New Intermediate Polar Cataclysmic Variable, Swift J183920.1—045350*. ApJ, 923, 243.
27. Champion, D., Cognard, I., Cruces, M., Desvignes, G., et al., 2020. *High-cadence observations and variable spin behaviour of magnetar Swift J1818.0-1607 after its outburst*. MNRAS, 498, 6044.

## Awards, Fellowships, & Grants \_\_\_\_\_

## AWARDS

2021 **Berman Award for Excellence in Experimental Physics**, The George Washington University

## FELLOWSHIPS

2023 **McWilliams Postdoctoral Fellowship**, Carnegie Mellon University

2016 **Davenport Research Fellowship**, Union College

2013-2017 **Presidential Scholarship**, Union College

## ACADEMIC HONOR SOCIETIES

2017 **Phi Beta Kappa**, Union College

2017 **Sigma Xi**, Union College

2016 **Sigma Pi Sigma**, Union College

2016 **The Order of Omega**, Union College

## GRANTS (PI AWARDS >\$390,000)

|      |                                                                                                                      |          |
|------|----------------------------------------------------------------------------------------------------------------------|----------|
| 2023 | <b>McWilliams/PSC Seed Grant (PI: B. O'Connor)</b> , <i>Renewing the connection between CMU and SALT</i>             | \$30,000 |
| 2023 | <b>Hubble Space Telescope Cycle 31 Award; GO-17583 (PI: B. O'Connor)</b> , Space Telescope Science Institute (STScI) | \$55,367 |
| 2023 | <b>Hubble Space Telescope Cycle 31 Award; GO-17492 (PI: B. O'Connor)</b> , Space Telescope Science Institute (STScI) | \$55,208 |
| 2023 | <b>Chandra Cycle 26 Award (PI: B. O'Connor)</b> , Smithsonian Astrophysical Observatory (SAO)                        | \$64,000 |
| 2023 | <b>Chandra Cycle 25 Award (PI: B. O'Connor)</b> , Smithsonian Astrophysical Observatory (SAO)                        | \$56,755 |
| 2022 | <b>Chandra Cycle 24 Award (PI: B. O'Connor)</b> , Smithsonian Astrophysical Observatory (SAO)                        | \$65,209 |
| 2021 | <b>Chandra Cycle 23 Award (PI: B. O'Connor)</b> , Smithsonian Astrophysical Observatory (SAO)                        | \$66,792 |

## Approved Telescope Proposals

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As of November 15, 2024: I am PI of 28 approved telescope proposals with a total financial award of \$390,000 since December 2021. I was involved in an additional 90 approved proposals as Co-I, for a total of 118 approved programs overall.

These proposals have been awarded time on the following telescopes and observatories:

*JWST, HST, Chandra, NuSTAR, XMM-Newton, NICER, Swift, Fermi, Gemini, Keck, LDT, LBT, GTC, SALT, SOAR, Subaru, DE-Cam, NEWFIRM, VLA, uGMRT, e-MERLIN, EVN, GBT.*

As PI I have been awarded 59 orbits (320 ks) of *HST*, 490 ks of *Chandra*, 54 hr (195 ks) of VLA, and 102 hr (367 ks) of Gemini.

## AS PRINCIPAL INVESTIGATOR (SORTED BY OBSERVATORY)

- HST Cycle 31**, Understanding the Hubble tension and jet physics through joint electromagnetic and gravitational wave observations of a neutron star merger – Awarded 28 orbits (2 ToO triggers of 5 epochs each) *GO-17583*
- HST Cycle 31**, Zooming in on the locations of short gamma-ray bursts – Awarded 31 orbits for 22 targets *GO-17492*
- Chandra Cycle 26**, An off-axis view of compact binary mergers – Awarded 250 ks of *Chandra* and 18 hr of VLA as a joint program *26400095*
- Chandra Cycle 25**, The collimation and energetics of short GRBs: searching for jet-breaks with *Chandra* – Awarded 80 ks (2 ToOs) *25400257*
- Chandra Cycle 24**, The collimation and energetics of short GRBs: searching for jet-breaks with *Chandra* – Awarded 80 ks (2 ToOs) *24400201*
- Chandra Cycle 23**, The collimation and energetics of short GRBs: searching for jet-breaks with *Chandra* – Awarded 80 ks (2 ToOs) *23400418*
- XMM-Newton Director's Discretionary Time**, Awarded 100 ks to study GRB 230307A.

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| 8.  | <b>NuSTAR Director's Discretionary Time</b> , Awarded 40 ks to study EP240408A.                                                                                |                 |
| 9.  | <b>VLA 2025A &amp; 2025B</b> , An off-axis view of compact binary mergers – Awarded 18 hr ToO                                                                  | VLA/25A-267     |
| 10. | <b>VLA 2024A &amp; 2024B</b> , An off-axis view of compact binary mergers – Awarded 18 hr ToO                                                                  | VLA/24A-320     |
| 11. | <b>SALT 2024-2</b> , SALT Spectroscopy of Rare Transients - Awarded 11 hr of ToO                                                                               | 2024-2-SCI-042  |
| 12. | <b>SALT 2024-1</b> , SALT Spectroscopy of Rare Transients - Awarded 11 hr of ToO                                                                               | 2024-1-SCI-042  |
| 13. | <b>Gemini-North 2024B</b> , Rapid Gemini and Hubble Space Telescope Observations of a kilonova – Awarded 12 hr of Rapid ToO                                    | GN-2024B-Q-136  |
| 14. | <b>Gemini-South 2024B</b> , Rapid Gemini and Hubble Space Telescope Observations of a kilonova – Awarded 20.5 hr of Rapid ToO                                  | GS-2024B-Q-132  |
| 15. | <b>Gemini-North 2024B</b> , Searching for gold: identification of the true counterpart via host galaxy spectroscopy – Awarded 8.5 hr of Standard ToO           | GN-2024B-Q-137  |
| 16. | <b>Gemini-South Director's Discretionary Time 2024A</b> , Unravelling the mystery of the fast X-ray transient EP240408A – Awarded 4.1 hr of Rapid ToO          | GS-2024A-DD-104 |
| 17. | <b>Gemini-South Director's Discretionary Time 2023A</b> , The distance and energetics of the second brightest GRB of all time – Awarded 3 hr of Rapid ToO      | GS-2023A-DD-104 |
| 18. | <b>Gemini-North 2023A</b> , Identifying the fingerprints of r-process heavy metals in a short GRB – Awarded 9.5 hr of Rapid ToO                                | GN-2023A-Q-131  |
| 19. | <b>Gemini-South 2023A</b> , Identifying the fingerprints of r-process heavy metals in a short GRB – Awarded 9.5 hr of Rapid ToO                                | GS-2023A-Q-130  |
| 20. | <b>Gemini-South Director's Discretionary Time 2022B</b> , Unveiling heavy elements from the ultra-long GRB 221009A – Awarded 3.5 hr of Rapid ToO               | GS-2022B-DD-104 |
| 21. | <b>Gemini-North 2022B</b> , Identifying the fingerprints of r-process heavy metals in a short GRB – Awarded 9.5 hr of Rapid ToO                                | GN-2022B-Q-130  |
| 22. | <b>Gemini-South 2022B</b> , Identifying the fingerprints of r-process heavy metals in a short GRB – Awarded 9.5 hr of Rapid ToO                                | GS-2022B-Q-134  |
| 23. | <b>Gemini-South 2022B</b> , Off-axis afterglows from compact binary mergers – Awarded 8.5 hr of Standard ToO                                                   | GS-2022B-Q-232  |
| 24. | <b>Gemini-South 2022A</b> , Identifying the fingerprints of r-process heavy metals in a short GRB – Awarded 9.5 hr of Rapid ToO                                | GS-2022A-Q-141  |
| 25. | <b>Gemini-South Director's Discretionary Time 2021B</b> , Probing the unusual long GRB 211227A with Gemini – Awarded 2.1 hr of Rapid ToO                       | DT-2021B-019    |
| 26. | <b>Lowell Discovery Telescope 2023A (Co-PI)</b> , Classically Scheduled Imaging and Spectroscopy of Transients and Their Host Galaxies – Awarded 5 full nights |                 |
| 27. | <b>Lowell Discovery Telescope 2022B (Co-PI)</b> , Classically Scheduled Imaging and Spectroscopy of Transients and Their Host Galaxies – Awarded 5 full nights |                 |
| 28. | <b>Lowell Discovery Telescope 2022A</b> , Gamma-ray bursts and their host environments – Awarded 4 half-nights                                                 |                 |

#### CO-I PROPOSALS (SORTED BY OBSERVATORY)

|    |                                                                                                                                    |             |
|----|------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1. | <b>JWST Cycle 3 (PI: L. Hu)</b> , An Archival Study of Cosmic Transients in Existing JWST Observations                             | GO-5965     |
| 2. | <b>JWST Cycle 2 (PI: E. Troja)</b> , Identifying the fingerprints of heavy r-process elements with the James Webb Telescope        | GO-3704     |
| 3. | <b>HST Cycle 32 (PI: E. Troja)</b> , Ultra-rapid observations of a gravitational wave source                                       | GO-17805    |
| 4. | <b>HST Cycle 31 (PI: M. Joyce)</b> , Far-UV detection of the low-mass companion responsible for Betelgeuse's long secondary period | GO/DD-17881 |
| 5. | <b>HST Cycle 31 (PI: E. Troja)</b> , A holistic view of compact binary mergers: from kilonova to afterglow                         | GO-17450    |

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| 6.  | <b>HST Cycle 30 (PI: E. Troja)</b> , The afterglow, supernova and distance scale of a record-breaking gamma-ray burst                                      | GO/DD-17298     |
| 7.  | <b>HST Cycle 30 (PI: E. Troja)</b> , Mapping the diversity of kilonovae through rapid Hubble observations of a short gamma-ray burst                       | GO-17175        |
| 8.  | <b>HST Cycle 29 (PI: E. Troja)</b> , Identifying the fingerprints of r-process heavy metals in a short GRB                                                 | GO-16846        |
| 9.  | <b>HST Cycle 25 (PI: E. Troja)</b> , Identify the signature of neutron star mergers through rapid Hubble observations of a short GRB                       | GO-15089        |
| 10. | <b>Gemini-South 2025A (PI: N. Klingler)</b> , Resolving the First Bow Shock Pulsar Wind Nebula in Near-IR                                                  | GS-2025A-Q-232  |
| 11. | <b>Gemini-South 2024B (PI: I. Andreoni)</b> , Gemini spectroscopy to identify the redshift of a peculiar fast X-ray transient discovered by Einstein Probe | GS-2024A-FT-113 |
| 12. | <b>Gemini-South 2023A (PI: S. Dichiara)</b> , Searching for the SN associated with the extremely bright GRB 230307A                                        | GS-2023A-DD-106 |
| 13. | <b>Gemini-South 2023A (PI: N. Klingler)</b> , Resolving the First Bow Shock Pulsar Wind Nebula in Near-IR                                                  | GS-2023A-Q-224  |
| 14. | <b>Gemini-North 2023A (PI: M. Im)</b> , Optical/NIR Follow-up Observation of Gravitational-Wave Sources                                                    | GN-2023A-Q-116  |
| 15. | <b>Gemini-South 2023A (PI: M. Im)</b> , Optical/NIR Follow-up Observation of Gravitational-Wave Sources                                                    | GS-2023A-Q-121  |
| 16. | <b>Gemini-North 2022B (PI: M. Im)</b> , Long-term Monitoring in Optical/NIR of Gravitational-wave Sources                                                  | GN-2022B-Q-117  |
| 17. | <b>Gemini-South 2022B (PI: M. Im)</b> , Long-term Monitoring in Optical/NIR of Gravitational-wave Sources                                                  | GS-2022B-Q-120  |
| 18. | <b>Gemini-North 2022B (PI: M. Im)</b> , Optical/NIR Follow-up Observation of Gravitational-Wave Sources                                                    | GN-2022B-Q-118  |
| 19. | <b>Gemini-South 2022B (PI: M. Im)</b> , Optical/NIR Follow-up Observation of Gravitational-Wave Sources                                                    | GS-2022B-Q-119  |
| 20. | <b>Gemini-South 2021A (PI: E. Troja)</b> , Mapping the diversity of neutron star mergers with rapid Gemini observations of short gamma-ray bursts          | GS-2021A-Q-102  |
| 21. | <b>Gemini-North 2021A (PI: E. Troja)</b> , Mapping the diversity of neutron star mergers with rapid Gemini observations of short gamma-ray bursts          | GN-2021A-Q-103  |
| 22. | <b>Gemini-North 2020B (PI: E. Troja)</b> , Mapping the diversity of neutron star mergers with rapid Gemini observations of short gamma-ray bursts          | GN-2020B-Q-102  |
| 23. | <b>Gemini-South 2020B (PI: E. Troja)</b> , Mapping the diversity of neutron star mergers with rapid Gemini observations of short gamma-ray bursts          | GS-2020B-Q-101  |
| 24. | <b>Keck 2024B (PI: I. Andreoni)</b> , Illuminating the r-process yield of neutron star mergers with Keck spectroscopy                                      |                 |
| 25. | <b>Keck 2023B (PI: S. B. Cenko)</b> , ToO Spectroscopy of GW Counterparts                                                                                  |                 |
| 26. | <b>Keck 2023A (PI: S. B. Cenko)</b> , ToO Spectroscopy of GW Counterparts                                                                                  |                 |
| 27. | <b>Keck 2022B (PI: S. B. Cenko)</b> , ToO Spectroscopy of GW Counterparts                                                                                  |                 |
| 28. | <b>SALT 2024-1 (PI: E. Dadiani)</b> , Spectral and Dynamical Analysis of a Dual AGN Candidate Using SALT RSS Spectroscopy                                  | 2024-1-SCI-041  |
| 29. | <b>Subaru 2024A (PI: T. Sakamoto)</b> , Short GRBs as a Key to Understand Population of Merging Neutron Stars                                              |                 |
| 30. | <b>Subaru 2023B (PI: T. Sakamoto)</b> , Short GRBs as a Key to Understand Population of Merging Neutron Stars                                              |                 |
| 31. | <b>Subaru 2023A (PI: T. Sakamoto)</b> , Short GRBs as a Key to Understand Population of Merging Neutron Stars                                              |                 |
| 32. | <b>GTC 2023A (PI: A. Watson)</b> , Characterizing Gravitational-Wave Mergers of Neutron Stars                                                              |                 |

33. **GTC 2022B (PI: A. Watson)**, Characterizing Gravitational-Wave Mergers of Neutron Stars
34. **Large Binocular Telescope 2023A (PI: Troja)**, Identifying the fingerprints of r-process heavy metals in a short GRB
35. **SOAR 2024B (PI: A. Palmese)**, Understanding compact binary formation and cosmology through gravitational wave host galaxies 2024B-730348
36. **DECam 2025A (PI: A. Palmese)**, DESI Transients Survey 2025A-729671
37. **DECam 2023B (PIs: I. Andreoni, A. Palmese)**, GW-MMADS: Gravitational Wave Multi-Messenger Astronomy DECam Survey 2023B-851374
38. **NEWFIRM 2025A (PI: D. Coulter)**, Finding and Characterizing Red and Distant Kilonovae with NEWFIRM 2025A-808226
39. **NEWFIRM 2024B (PI: D. Coulter)**, Finding and Characterizing Red and Distant Kilonovae with NEWFIRM 2024B-945577
40. **NEWFIRM 2024A (PI: D. Coulter)**, Finding and Characterizing Red and Distant Kilonovae with NEWFIRM 2024A-706757
41. **Lowell Discovery Telescope 2023A (PI: S. B. Cenko)**, Target of Opportunity Transient Follow-Up with LDT
42. **Lowell Discovery Telescope 2023A (PI: I. Andreoni)**, ToO Observations of Gravitational Wave Counterparts in the Fourth LIGO-Virgo-KAGRA Observing Run
43. **Lowell Discovery Telescope 2022B (PI: S. B. Cenko)**, Target of Opportunity Transient Follow-Up with LDT
44. **Lowell Discovery Telescope 2022B (PI: I. Andreoni)**, ToO Observations of Gravitational Wave Counterparts in the Fourth LIGO-Virgo-KAGRA Observing Run
45. **Lowell Discovery Telescope 2022A (PI: A. Gottlieb)**, LDT observations of Fast Radio Bursts: counterparts and environment
46. **Lowell Discovery Telescope 2022A (PI: Cenko)**, Target of Opportunity Gamma-Ray Burst Follow-Up with LDT
47. **Lowell Discovery Telescope 2021B (PI: Cenko)**, Target of Opportunity Gamma-Ray Burst Follow-Up with LDT
48. **Lowell Discovery Telescope 2021B (PI: S. Dichiara)**, Gamma-ray bursts and their host environments
49. **Lowell Discovery Telescope 2021B (PI: E. Troja)**, LDT observations of Fast Radio Bursts: counterparts and environment
50. **Lowell Discovery Telescope 2021A (PI: S. Dichiara)**, Gamma-ray bursts and their host environments
51. **Lowell Discovery Telescope 2021A (PI: E. Troja)**, LDT observations of Fast Radio Bursts: counterparts and environment
52. **Fermi Cycle 14 (PI: C. Kouveliotou)**, Magnetar Observations with the Fermi/Gamma Ray Burst Monitor
53. **Chandra Cycle 26 Award (PI: E. Troja)**, Identifying the fingerprints of r-process heavy metals in a nearby GRB – Awarded *Chandra*, *JWST*, and VLA as a joint program 26500416
54. **Chandra Cycle 25 (PI: E. Troja)**, Beyond the GRB jet: searching for the remnant of a neutron star merger
55. **Chandra Cycle 25 (PI: E. Troja)**, GOTCHA! Gravitational wave counterparts Observed with CHAndra
56. **Chandra Cycle 24 (PI: E. Troja)**, GOTCHA! Gravitational wave counterparts Observed with CHAndra
57. **Chandra Cycle 24 (PI: E. Troja)**, Identifying the fingerprints of r-process heavy metals in a short GRB
58. **Chandra Cycle 24 (PI: S. Dichiara)**, Chandra Sub-arcsecond Localization of Swift Short GRBs
59. **Chandra Cycle 23 (PI: C. Kouveliotou)**, *Chandra* ToO observations of Phase II *Swift* Deep Galactic Plane Survey (DGPS) sources

60. **Chandra Cycle 23 (PI: E. Troja)**, Beyond the GRB jet: searching for the remnant of a neutron star merger
61. **Chandra Cycle 23 (PI: E. Troja)**, Identifying the fingerprints of r-process heavy metals in a short GRB
62. **Chandra Cycle 23 (PI: S. Dichiara)**, Chandra Sub-arcsecond Localization of Swift Short GRBs
63. **Chandra Cycle 22 (PI: E. Troja)**, The Collimation and Energetics of Short GRBs: Searching for Jet-breaks with Chandra
64. **Chandra Director's Discretionary Time (PI: A. O'Grady)**, X-ray detection of the low-mass companion responsible for Betelgeuse's long secondary period
65. **Chandra Director's Discretionary Time (PI: E. Troja)**, A luminous kilonova or a faint supernova? The curious case of GRB210704A
66. **Chandra Director's Discretionary Time (PI: L. Piro)**, Unraveling the nature of the persistent radio source associated to FRB201124A with *Chandra*
67. **Swift Cycle 20 (PI: S. Dichiara)**, Searching High and Low for Elusive Short GRBs
68. **Swift Cycle 18 (PI: S. Dichiara)**, Searching High and Low for Elusive Short GRBs
69. **XMM-Newton AO21 (PI: E. Troja)**, The collimation and energetics of short GRBs: searching for jet-breaks with XMM
70. **XMM-Newton AO20 (PI: E. Troja)**, Identifying the fingerprints of r-process heavy metals in a short GRB
71. **NuSTAR Director's Discretionary Time (PI: C. Kouveliotou)**, *Swift* Galactic Plane Survey Key Project – Utilized 9 *NuSTAR* DDTs of Galactic X-ray sources identified through the *Swift* Deep Galactic Plane Survey. The Survey was a *NuSTAR* Legacy Survey Program through 2019.
72. **NICER Cycle 4 (PI: N. Klingler)**, The transitional millisecond pulsar candidate 4FGL J1943.9+2841
73. **NICER Cycle 4 (PI: C. Kouveliotou)**, *NICER* ToO observations of *Swift*/XRT Deep Galactic Plane Survey (DGPS) sources
74. **NICER Cycle 3 (PI: C. Kouveliotou)**, *NICER* ToO observations of *Swift*/XRT Deep Galactic Plane Survey (DGPS) sources
75. **EVN E23 (PI: G. Bruni)**, Characterising the progenitors of fast radio bursts with the EVN
76. **EVN E21 (PI: G. Bruni)**, Characterising the progenitors of fast radio bursts with the EVN
77. **EVN DDT (PI: F. Panessa)**, Disclosing the nature of the persistent radio source associated to FRB20201124A
78. **e-MERLIN Cycle 13 (PI: G. Bruni)**, Characterising the progenitors of fast radio bursts with e-MERLIN
79. **e-MERLIN DDT (PI: G. Bruni)**, Disclosing the nature of the persistent radio source associated to FRB 20201124A with e-MERLIN
80. **GMRT Cycle 47 (PI: E. Troja)**, Gravitational Wave events: radio follow-up with uGMRT
81. **GMRT Cycle 42 (PI: G. Bruni)**, Spectral characterization of the persistent radio emission in fast radio bursts
82. **GMRT DDT (PI: G. Bruni)**, Characterising starburst activity in the host of the repeating FRB 20201124A
83. **VLA 2025A (PI: E. Troja)**, The collimation and energetics of short gamma-ray bursts
84. **VLA 2024A (PI: E. Troja)**, The collimation and energetics of short gamma-ray bursts
85. **VLA 2022B (PI: E. Troja)**, The collimation and energetics of short gamma-ray bursts
86. **VLA 2022B (PI: S. Chastain)**, Electromagnetic counterparts of gravitational wave events
87. **VLA 2021B (PI: E. Troja)**, The collimation and energetics of short gamma-ray bursts
88. **VLA 2021B (PI: E. Troja)**, Beyond the GRB jet: searching for the remnant of a neutron star merger
89. **VLA 2021A (PI: E. Troja)**, Beyond the GRB jet: searching for the remnant of a neutron star merger
90. **GBT 2025A (PI: N. Klingler)**, GBT Observations of a Candidate Transitional Millisecond Pulsar

EB099

EB094

GBT25A-273



## Presentations

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### INVITED DISCUSSION PANELS

1. **Dawn of Gravitational Wave Astronomy and Astrophysics (Dawn VII) Meeting.** Invited speaker and panelist. *Multi-messenger Astrophysics in the A# Era*. Vancouver, Canada. June 2024. Contributed to white paper (McIver et al.).
2. **Gravitational Wave Physics and Astronomy Workshop (GWPAW).** Discussion panelist. *What Astronomers need to do MMA*. Birmingham, UK. May 2024.

### INVITED TALKS

1. **GRB+CE Workshop 2024.** Invited talk (20m). *Constraints On the Population of Off-axis Short Gamma-ray Bursts*. Playa del Carmen, Mexico. December 2024.
2. **COSPAR 2024.** Invited talk (20m). *A search for hostless short GRBs with large aperture telescopes*. Busan, South Korea. July 2024.
3. **GROWTH MMA Meeting.** Invited talk (15m). *A structured jet explains the extreme GRB 221009A*. Presented virtually. March 2023.
4. **Union College.** Invited colloquia (45m). *The Transient Universe: Compact Objects Near and Far*. Schenectady, NY. February 2023.
5. **Harvard-Smithsonian Center for Astrophysics (CfA).** Invited talk (45m) at High Energy Seminar. *The Transient Universe: Compact Objects Near and Far*. Cambridge, MA. November 2022.
6. **Lowell Observatory.** Invited talk (15m) at A Decade of Exploration with the Lowell Discovery Telescope. *The host galaxies and environments of short gamma-ray bursts*. Presented virtually. October 2022.
7. **California Institute of Technology (Caltech).** Invited talk (45m) at Astronomy Tea Talk series. *The Transient Universe: Compact Objects Near and Far*. Presented virtually. October 2022.
8. **University of California, Berkeley.** Invited talk (45m) at Explosive Astro talk series. *The Transient Universe: Compact Objects Near and Far*. Berkeley, CA. September 2022.
9. **University of California, Santa Cruz.** Invited talk (45m) at FLASH Seminar. *The Transient Universe: Compact Objects Near and Far*. Santa Cruz, CA. September 2022.
10. **Universidad Nacional Autónoma de México.** Invited talk (45m) at High Energy Astrophysics (HEAP) seminar. *Shedding light on hostless short GRBs with large aperture telescopes*. Presented virtually. March 2022.

### CONTRIBUTED PRESENTATIONS

1. **Dawn VII Satellite Workshop: Multi-Messenger Astrophysics with GWs.** Contributed talk (15m). *Constraints On the Population of Off-axis Short Gamma-ray Bursts*. Vancouver, CA. June 2024.
2. **Gravitational Wave Physics and Astronomy Workshop (GWPAW).** Contributed talk (15m). *Constraints On the Population of Off-axis Short Gamma-ray Bursts*. Birmingham, UK. May 2024.
3. **GRB50 Meeting.** Contributed talk (10m). *A structured jet explains the extreme GRB 221009A*. Warrenton, VA. August 2023.
4. **High Energy Astrophysics Division (HEAD 20) Meeting.** Dissertation talk (15m). *The Transient Universe: Compact Objects Near and Far*. Kona, HI. March 2023.
5. **241st meeting of the American Astronomical Society (AAS).** Dissertation talk (15m). *The Transient Universe: Compact Objects Near and Far*. Seattle, WA. January 2023.
6. **Kilonova: Multimessenger and Multiphysics.** Contributed Early Career talk (20m) at WE-Heraeus Seminar. *The host galaxies and environments of short gamma-ray bursts*. Bad Honnef, Germany. November 2022.
7. **Lowell Discovery Telescope Partners' Meeting at Boston University.** Contributed talk (15m). *The host galaxies and environments of short gamma-ray bursts*. Presented virtually. November 2022.
8. **IAU Symposium 363.** Contributed talk (20m). *Shedding light on hostless short GRBs with large aperture telescopes*. Presented virtually. December 2021.
9. **Marcel Grossman 16th Meeting.** Contributed talk (15m). *Constraints on kilonova emission in two short GRBs at  $z \sim 0.5$* . Presented virtually. June 2021.
10. **European Astronomical Society (EAS) Annual Meeting.** Contributed talk (15m). *Constraints on kilonova emission in two short GRBs at  $z \sim 0.5$* . Presented virtually. June 2021.
11. **Square Kilometer Array (SKA) Science Conference.** Contributed talk (10m). *Constraints on kilonova emission in two short GRBs at  $z \sim 0.5$* . Presented virtually. March 2021.
12. **237th meeting of the American Astronomical Society (AAS).** Contributed talk (5m). *Constraints on kilonova emission in two short GRBs at  $z \sim 0.5$* . Presented virtually. January 2021.
13. **Chandra Frontiers in Time Domain Astrophysics.** Contributed talk (15m). *The merger environments of short gamma-ray bursts*. Presented virtually. October 2020.

## Press \_\_\_\_\_

To date, my research has generated >200 press releases. A few selected releases are listed below.

### The “Brightest of All Time” Gamma-Ray Burst.

- The Science Advances [article](#) on GRB 221009A led to 123 news reports, below are some highlights.
- GWU: <https://gwtoday.gwu.edu/what-makes-gamma-ray-burst-brightest-all-time>
- NuSTAR: <https://www.nustar.caltech.edu/news/nustar230608>
- NOIRLab: <https://noirlab.edu/public/blog/the-brightest-of-all-time/>
- Nature Italy: <https://www.nature.com/articles/d43978-023-00084-x>
- Independent: [Independent](#)
- Forbes: [Forbes](#)

### Kilonova Discovery Challenges our Understanding of Gamma-Ray Bursts.

- Nature “Behind-the-paper”: <https://astronomycommunity.nature.com/>
- NOIRLab: <https://noirlab.edu/public/news/noirlab2228/>
- Inverse: [Inverse](#)
- NASA: [NASA](#)
- GWU: [GWU](#)

### Record-Breaking Gamma-Ray Burst Possibly Most Powerful Explosion Ever Recorded.

- NOIRLab: <https://noirlab.edu/public/news/noirlab2224/>
- NSF Science Now: <https://youtu.be/Do2oFQjAS8o>
- Space.com: [Space.com](#)
- Forbes: [Forbes](#)
- CNN: [CNN](#)

### Gemini Telescopes Help Uncover Origins of Castaway Gamma-Ray Bursts.

- NOIRLab: <https://noirlab.edu/public/news/noirlab2218/>
- Keck: <https://www.keckobservatory.org/castaway-grbs/>
- UMD: <https://cmns.umd.edu/news-events/features/4958>

## Teaching Experience \_\_\_\_\_

2019    **Secondary Instructor**, Solar System Astronomy, The George Washington University  
2017-2019    **Graduate Teaching Assistant**, Physics and Astronomy, The George Washington University

## Mentoring Experience \_\_\_\_\_

### STUDENT SUPERVISION

To date I have directly supervised or co-supervised 4 graduate students and 7 undergraduate students. These are students whom with I have met with on a weekly basis to discuss their ongoing research projects.

\*Resulted in a publication. \*\*Publication in preparation.

|                                                                                               |                        |
|-----------------------------------------------------------------------------------------------|------------------------|
| <b>Co-supervised a graduate student</b> , Xander Hall, Carnegie Mellon University             | Sep 2024 -<br>now      |
| <b>Co-supervised an undergraduate student</b> , Reed Barmada**, Carnegie Mellon University    | Sep 2024 -<br>now      |
| <b>Co-supervised an undergraduate student</b> , Frank Price, Carnegie Mellon University       | Sep 2024 -<br>now      |
| <b>Co-supervised a graduate student</b> , Hannah Skobe, Carnegie Mellon University            | Jun 2024 -<br>now      |
| <b>Supervised an undergraduate student</b> , Jackson Chen**, Carnegie Mellon University       | Jun 2024 -<br>now      |
| <b>Supervised an undergraduate student</b> , Samanvita Singhania, Carnegie Mellon University  | Jun 2024 - Dec<br>2024 |
| <b>Supervised an undergraduate student</b> , Dylan Rossi, Carnegie Mellon University          | Jan 2024 - Aug<br>2024 |
| <b>Co-supervised an undergraduate student</b> , Rav Kaur*, University of California, Berkeley | Oct 2023 - Sep<br>2024 |
| <b>Co-supervised an undergraduate student</b> , Aidan Catalano, Carnegie Mellon University    | Summer<br>2023/2024    |
| <b>Supervised a graduate student</b> , Seth Gagnon*, The George Washington University         | Summer 2022            |
| <b>Supervised a graduate student</b> , Alex van Kooten*, The George Washington University     | Summer 2021            |

## STUDENT MENTORING

In addition to direct student supervision, I have mentored multiple graduate students, aiding in writing and editing applications and proposals, and assisting with their research projects.

|                                                                                                    |                   |
|----------------------------------------------------------------------------------------------------|-------------------|
| <b>Mentored a graduate student</b> , Malte Busmann*, Ludwig Maximilian University                  | Nov 2023 -<br>now |
| <b>Mentored a graduate student</b> , Ekaterine Dadiani, Carnegie Mellon University                 | Nov 2023 -<br>now |
| <b>Mentored a graduate student</b> , Tomás Cabrera*, Carnegie Mellon University                    | Sep 2023 -<br>now |
| <b>Mentored a graduate student</b> , Keerthi Kunnumkai**, Carnegie Mellon University               | Sep 2023 -<br>now |
| <b>Mentored a graduate student</b> , Gokul Srinivasaragavan*, University of Maryland, College park | Sep 2022 -<br>now |

## Outreach & Professional Development

### CONFERENCE ORGANIZATION

2024 **SOC Member**, *Swift* Senior Review Workshop

### PEER REVIEW

2024 **Proposal reviewer**, *James Webb Space Telescope (JWST)* Guest Observer Peer Review

2024 **Proposal reviewer**, NASA Astrophysics Data Analysis Program (ADAP)

2024 **TAC Member**, *Chandra* Peer Review

2024 - now **TAC Member**, NOIRLab Time Allocation Committee (TAC)

2024 - now **Journal referee**, The Monthly Notices of the Royal Astronomical Society (MNRAS)  
 2024 - now **Proposal reviewer**, *James Webb Space Telescope (JWST)* Director's Discretionary Time  
 2024 **Proposal reviewer**, Israel Science Foundation  
 2024 - now **Co-chair**, Southern African Large Telescope TAC for Carnegie Mellon University  
 2023/2024 **Member**, McWilliams Fellowship Hiring Committee, Carnegie Mellon University  
 2023 - now **Journal referee**, Nature Astronomy  
 2023 - now **Journal referee**, Astronomy & Astrophysics (A&A)  
 2022 - now **Journal referee**, The Astrophysical Journal (ApJ)  
 2023 **Proposal reviewer**, Subaru Telescope Time Allocation Committee  
 2023 **Proposal reviewer**, Italian Time Allocation Committee for TNG/REM  
 2022 **Proposal reviewer**, Gemini Observatory Canadian Time Allocation Committee (CanTAC)

## PROFESSIONAL MEMBERSHIPS

2024 - now **Nancy Grace Roman WFI Working Group**, Member  
 2024 - now **LSST Transients and Variable Stars (TVS) Science Collaboration**, Member  
 2023 - now **Gravitational Wave Multi-Messenger Astronomy DECam Survey (GW-MMADS)**, Member/Observer  
 2023 - now **DECam Alliance for Transients (DECAT)**, Member/Observer  
 2023 - now **DECam Survey of Intermediate Redshift Transients (DESIRT)**, Member  
 2023 - now **The Dark Energy Spectroscopic Instrument (DESI)**, Junior Member  
 2023 - now **The Gravity Collective**, Member  
 2023 - now **HEASARC Users Group (HUG)**, Member  
 2023 - now **Athena Science Working Group (SWG3.6: Athena multimessenger)**, Member  
 2022 - 2024 **STROBE-X Science Working Group**, Member  
 2020 - 2023 **Gamow Explorer Science Team**, Member  
 2020 - 2023 **Swift Deep Galactic Plane Survey (DGPS)**, Observation Lead  
 2020 - 2023 **MeerKAT Galactic Plane Survey**, Member

## WEEKLY COLLOQUIA

2024 - now **McWilliams Center Astronomy Seminar Series**, Organizer  
 2023 - now **CMU Multi-Messenger Astronomy Meetings**, Organizer/Presenter  
 2023 - now **CMU Astronomy Journal Club**, Member/Presenter  
 2022 - 2023 **GWU Astronomy Data Analysis Seminars**, Organizer/Presenter  
 2022 - 2023 **UMD Transient Astronomy Meetings**, Member/Presenter  
 2019 - 2023 **NASA GSFC GRB Lunch**, Member/Presenter  
 2018 - 2023 **GWU Astronomy Group Meetings**, Member/Presenter

## SERVICE AND OUTREACH

|                    |                                                                                             |                |
|--------------------|---------------------------------------------------------------------------------------------|----------------|
| 2024               | <b>Astronomy on Tap</b> , Speaker                                                           | Pittsburgh, PA |
| 2022               | <b>2022 Physics Congress (PhysCon)</b> , Volunteer                                          | Washington, DC |
| 2018 & 2019        | <b>Astronomy Festival on the National Mall</b> , Volunteer                                  | Washington, DC |
| 2016-2017          | <b>Union College Student Affairs Council</b> , Student Representative                       | Schectady, NY  |
| 2016-2017          | <b>Union College Student Conduct Committee</b> , Committee Member                           | Schectady, NY  |
| 2016 & 2017        | <b>Dudley Observatory at Museum of Innovation and Science</b> , Volunteer at Astronomy Days | Schectady, NY  |
| 2015-2016          | <b>Union College Men's Club Soccer</b> , Treasurer                                          | Schectady, NY  |
| 2015 & 2016        | <b>Special Olympics New York Annual 5k rUndead Event Service</b> , Volunteer                | Schectady, NY  |
| 2014, 2015, & 2016 | <b>John Calvin Toll Day of Community Service</b> , Volunteer                                | Schectady, NY  |